

Cluster- and Reach-scalable Optical Switching for Scale-across AI System

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Abstract: We propose an OCS-based scale-across architecture with reach- and cluster-scalability, while reducing the number of deployed cables. Experimental results demonstrate that 30-km extending between clusters does not deteriorate AI job completion time.

1. Introduction

With the rapid scaling of large AI models [1], the number of GPUs required is continuously increasing, making it challenging to realize AI infrastructure within a single data center—particularly due to limitations in power supply, thermal management, and physical space. To overcome these constraints and enable scalable GPU availability, interconnection across multiple data centers, i.e., scale-across [2], has emerged as a critical architectural approach.

As communication distances increase, scale-across architectures typically rely on metro optical networks. However, due to the unique characteristics of AI workloads, this scenario diverges from existing research on optically interconnected distributed data centers. In prior studies, optical transport network (OTN) [3, 4] formats are commonly employed to carry inter-data center traffic, with wavelength routing and switching implemented via reconfigurable optical add-drop multiplexers (ROADMs). This approach presents several limitations: (1) High-order formats like OTU3/OTU4 [5] lack transparency for GPU traffic, and shared resources with legacy services lead to congestion and unpredictable latency. (2) Conventional ROADMs suffer from limited connectivity due to constraints in degree and port counts of a wavelength selective switch (WSS). In scenarios involving large-scale GPU cluster interconnection, this can lead to switchover delays on the order of minutes or even service blocking, thereby restricting the scalability of scale-across deployments [6]. (3) AI workloads show complex communication patterns that require dynamic logical topologies. Metro network topologies based on OTN are typically ring-based or constrained by pre-deployed fiber infrastructure, making it difficult to dynamically adapt logical topologies to AI traffic patterns.

To address the network requirements of AI workloads—including guaranteed bandwidth, low and predictable latency, high connectivity, and support for complex traffic patterns—we propose a scale-across optical gateway based on wavelength-selective optical circuit switching (OCS), extended from the scale-up side. (1) By directly interfacing with scale-up switches via ZR pluggable modules (ZR), the gateway enables RDMA-over-WDM capabilities, reducing GPU communication overhead and mitigating congestion, latency jitter, and throughput instability associated with traditional TCP/IP protocols. The maximum deployment distance between any two adjacent GPU clusters can reach up to 80 km due to ZR reachability. (2) Utilizing optical switching and multi-port WSS reduces latency in large-scale scenarios and increases scalability. Additionally, employing SDM and WDM achieves a logically fully-connected topology, while reducing the amount of fiber required for fully-connected connection scenarios. (3) We enable GPU traffic to be mapped onto a specific optical path, achieving dynamic adaptation to the AI communication patterns. This reduces the potential for congestion and packet loss over long-distance networks. Subsequently, this paper analyzes the fiber-saving advantage and cluster-scalability of scale-across topologies interconnected through these gateways. To show the reach-scalability, the transmission characteristics of the proposed optical gateway are also evaluated. Furthermore, we evaluate the performance impact of scale-across GPU communication on AI jobs using our constructed platform. We successfully interconnected four GPU clusters over 30 km each. Experimental results show that AI job completion times are on par with those of equivalent-scale GPU clusters within a single data center.

2. Optical Gateway and Scale-across Architecture

The proposed scale-across architecture adopts a three-tier topology. As illustrated in Fig. 1 (a), the cabling topology conforms to a subgraph of a 2D-lattice, with each cluster node having a maximum degree of four. Each edge in the cabling topology comprises parallel links, which can be physically implemented using bundles of single-mode fibers (SMFs) or M-core multi-core fibers (MCFs). Given the heterogeneous number of clusters in practical

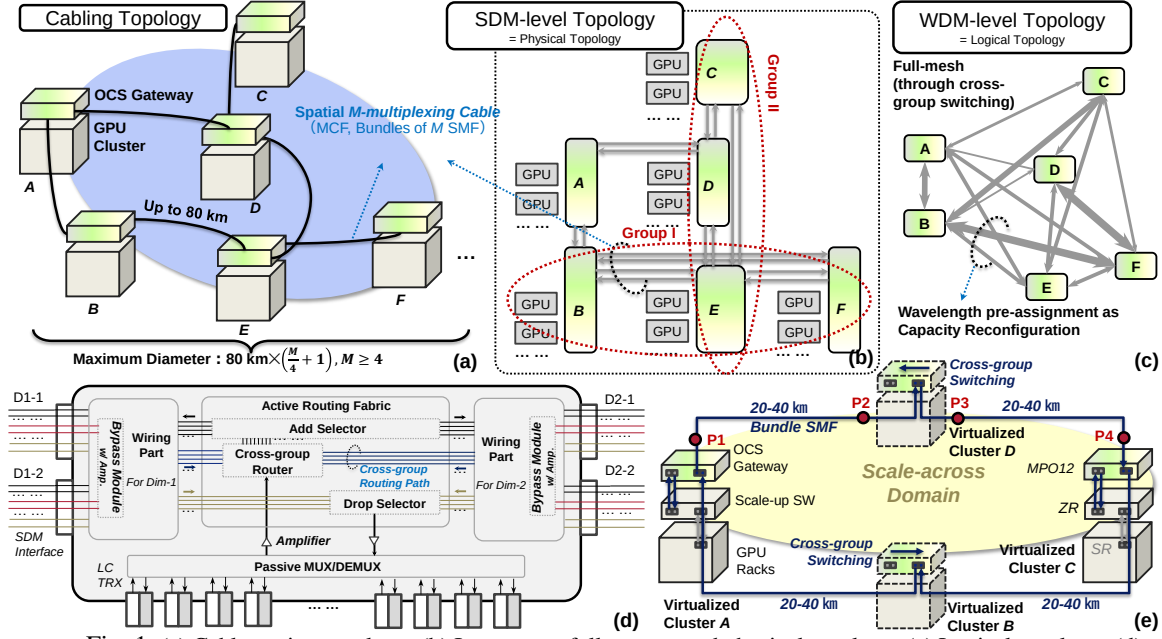


Fig. 1. (a) Cable-saving topology; (b) Intra-group fully-connected physical topology; (c) Logical topology; (d) Functional compositions of the optical gateway; (e) Experimental setup.

deployments, we partition the clusters into multiple groups, as shown in Fig. 1 (b). Each corresponding to a segment along either the vertical or horizontal dimension of a 2D-lattice. Fully-connected shape is maintained within each group, and a cluster may belong to up to two groups. When a cluster (e.g., *E*) is part of two groups, it serves as a WDM-layer cross-group switching node: e.g., wavelengths originating from a source node *B* in *Group I* are re-selected via node *E* and forwarded to a destination node *C* in *Group II*. In this method, any two clusters are connected via at most one cross-group switching, thereby achieving fully-connected topology at the WDM layer. Furthermore, as shown in Fig. 1 (c), by adjusting the number of wavelengths—corresponding to the number of ZRs—the link capacity can be flexibly adapted to meet the throughput demands of AI job communication patterns. This dynamic adjustment does not interrupt the data link layer, ensuring seamless inter-cluster communications.

The optical gateway exhibits reach-scalability comparable to the specifications defined by Open ZR [7] supporting interconnect distances of up to 80 km. Considering to use M-SDM cables, $M/4 + 2$ nodes can be fully-connected within a single group. Given that inter-group switching introduces OSNR degradation, we conservatively limit the architecture to a maximum of one cross-group switching. Even under this constraint, arbitrary interconnection among clusters comprising up to $(M/4 + 2)^2$ nodes remains feasible. The architecture achieves a maximum diameter of $(M/4 + 1) \times 80$ km. For example, using MPO12 as the SDM interfaces, an interconnect diameter of up to 320 km can be realized.

To realize the highly scalable and flexible scale-across architecture, the key lies in designing the optical gateway with the appropriate functions, as illustrated in Fig. 1 (d). The gateway employs passive MUX/DEMUX components to facilitate the add/drop of ZR wavelengths, supporting more than 32 channels. An active routing fabric is responsible for wavelength routing within and across groups, and can be implemented using various technologies such as WSS, arrayed waveguide gratings (AWGR) combined with cross-bar switches, or micro-ring resonator arrays [8]. This paper presents a concrete implementation example based on four WSS modules. Two of the WSSs are used to route wavelengths to interfaces corresponding to different dimensions (groups), or to select wavelengths arriving from those dimensions. The other two WSSs couple the filtered wavelengths from one group into another, thereby enabling cross-group switching. In wiring parts for different dimensions, the specific cores of the SDM interfaces are cross-connected, and optical amplifiers are incorporated, which together enable direct optical paths across the gateway node within the same group.

3. Performance Evaluation

The setup is shown in Fig. 1 (e). In either *Virtualized Cluster A* or *C*, each cluster is equipped with two Nvidia H100 GPUs and their corresponding two network interface cards (NICs). Each NIC is connected to a scale-up RDMA-over-Converged Ethernet (RoCE) switch via 400 G SR pluggable modules. Each RoCE switch is connected to the optical gateway through 400 G ZRs. The wavelength of a ZR is pre-configured to match the MUX/DEMUX ports of the optical gateway. During execution of the system, the configured wavelength is always fixed. To evaluate the impact of cross-group switching to the scale-across network, we use the other two virtualized clusters without GPUs to perform only functions of optical gateways. The SDM interfaces used to connect the optical gateways

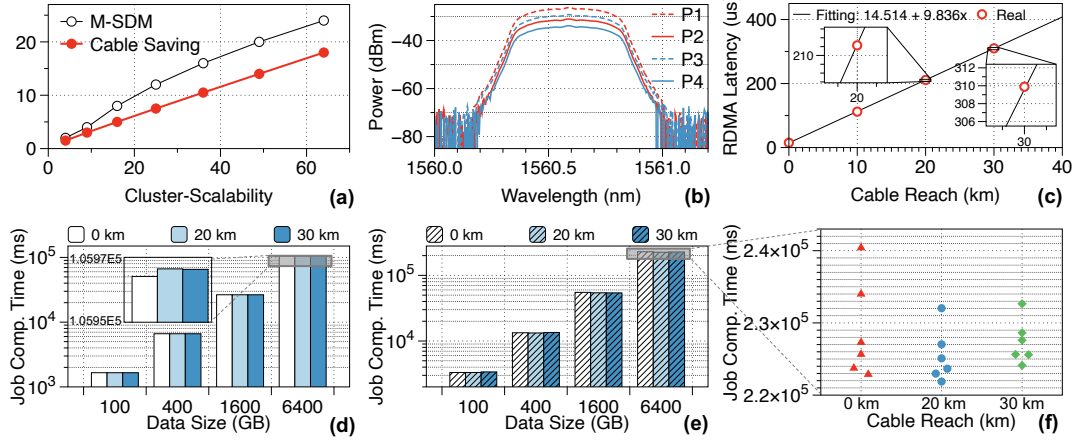


Fig. 2. (a) M-SDM requirement and cable-saving ratio vs. number of clusters in scale-across topology; (b) Spectra monitored across the optical gateway; (c) RDMA latency tests; (d) Job completion time of All-to-All pattern and (e) job completion time of Bi-direction Ring AllReduce pattern with various fiber extensions vs. training data size; (f) Job completion time jitter of Bi-direction Ring AllReduce pattern.

are MPO12. We additionally employ MPO-LC fan-out cables to extend the SMFs reach between two adjacent clusters, spanning 20 km and 30 km, respectively. This can emulate bundles of SMFs, as a type of SDM cables.

We first monitored the optical power profile at four points [through $P1$ to $P4$ shown in Fig. 1 (e)] in the scenario of 20-km fiber extension, showing the effects of amplifiers inside the optical gateways in Fig. 2 (b). The output power of a ZR is typically -8 dBm. Although OSNR cannot be directly measured, it can be inferred that cross-grouping leads to OSNR degradation, which constitutes a primary limitation on scalability imposed by transmission characteristics. In addition, the optical gateway used in the experiment is equipped with a mini-EDFA whose amplification capability is limited, thereby preventing extension tests beyond 30 km. Enhancing the amplification capacity of the EDFA could further improve the reach-scalability.

Subsequently, we run RDMA flows and AI jobs with communication patterns of *All-to-All* and *Bi-direction Ring AllReduce*. As shown in Fig. 2 (c), RDMA latency measurements indicate that as the length of SDM optical cables increases, end-to-end latency exhibits a linear growth trend. This significantly amplifies the volume of in-flight traffic within the network, thereby introducing substantial risks of congestion and packet loss. To mitigate these risks, we adopt a one-to-one mapping strategy, assigning each GPU communication flow to a dedicated wavelength-specific optical path. This approach effectively eliminates contention and collision among flows, ensuring deterministic bandwidth allocation and stable transmission performance. Therefore, the job completion times are minimally affected by increased fiber distances, with deviations ranging from several tens to a few hundreds of milliseconds, regardless of communication patterns, as shown in both Fig. 2 (d) and (e). A further comparison between Fig. 2 (e) and (f) indicates that these deviations are not primarily due to increased propagation delay. Instead, they are predominantly attributed to uncertainties at the software layer, which contribute significantly greater jitter than the impact introduced by fiber extension.

4. Conclusion

We proposed an OCS gateway-based scale-across architecture that enables both reach- and cluster-scalability. By leveraging WDM and SDM, the design forms a logically fully-connected topology among N geo-distributed clusters with configurable link capacity, while reducing cable deployment from $O(N^2)$ to $O(N)$. By leveraging ZR modules between gateways and scale-up switches, we achieve RDMA-over-WDM and reach-scalability for metro-scale GPU interconnection. Under this design, we interconnected four GPU clusters over 30 km each and conducted distributed AI training over them. Experimental results show that, under this design, AI job completion times are on par with those of equivalent-scale GPU clusters within a single data center.

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